# Documentation for Store Sales Dataset Analysis Project:

## 1) Project Planning & Management: (خطة إدارة المشروع)

### Project Idea: (فكرة المشروع)

Analyzing the Carrefour "store sales dataset from 2013 to 2023" to extract valuable business and marketing insights.

### Objectives: (أهداف المشروع)

- Build a robust data model and perform data cleaning and preprocessing.
- Identify and answer key analytical questions for decision-makers (e.g., the impact of product categories and regions on sales performance).
- Forecast future sales and trends using statistical and machine learning models.
- Build an interactive visualization dashboard summarizing the project's findings.

### Project Scope: (نطاق المشروع)

The project focuses exclusively on analyzing the Carrefour_new_2013_to_2023_products .xlsx dataset using SQL, Python (Pandas, Matplotlib, Scikit-learn), and Tableau.

### Timeline & Milestones: (الخطة الزمنية)

- **Week 1:** Build Data Model, Data Cleaning, and Preprocessing. (Deliverables: Cleaned dataset, data preprocessing notebook).
- **Week 2:** Analysis Questions Phase using SQL and Python. (Deliverables: Set of answered analysis questions).
- **Week 3:** Forecasting Questions Phase using Python (scikit-learn). (Deliverables: Visualization plots answering forecasting questions).
- **Week 4:** Visualization Dashboard and Final Presentation using Tableau. (Deliverables: Visualization dashboard, final report, and presentation).

### Roles & Responsibilities: (توزيع المهام)

- Ali Farag / Amr Khaled: Responsible for Week 1 tasks (Data modeling, cleaning using Python).
- Mariam Esam: Responsible for Week 2 tasks (Formulating analysis questions).
- Hassan Hammad: Responsible for Week 3 tasks (Building forecasting models using scikit-learn).
- Heba Mohamed: Responsible for Week 4 tasks (building the Tableau dashboard and make Presentation).

**Risk Management: (المخاطر المحتملة)**

- **Risk:** High percentage of missing values or inconsistent data formats in older records (e.g., 2013).
- **Mitigation:** Apply appropriate data imputation and handling techniques using Python (Pandas) during the preprocessing phase to ensure data integrity.

**Success Metrics: (مؤشرات نجاح المشروع)**
Successfully delivering a clean dataset, providing accurate answers to business questions, generating reliable forecasting plots, and deploying a fully functional, user-friendly Tableau dashboard.

## 2) Literature Review: (المشاريع المشابهة)

**Similar Project:**
There are numerous similar projects in the retail analytics domain, such as Walmart Sales Forecasting, Market Basket Analysis for large supermarkets, and retail demand prediction models.

**Comparison & Project Value: (المقارنة وتوضيح الاختلاف)**

- **Limitations of other projects:** Many existing projects rely on highly aggregated data or cover very short timeframes, lacking granular details about product categories over a long period.
- Our Project's Edge: Our project utilizes a decade of real-world data (10 years: 2013-2023) from Carrefour. This extended timeframe allows for highly accurate detection of seasonality and long-term trends. Furthermore, integrating advanced machine learning libraries like scikit-learn for forecasting, combined with Tableau for visualization, provides a more comprehensive and actionable tool for decision-makers.

## 3) Requirements Gathering: (المتطلبات)

**Stakeholders: (الأطراف المستفيدة من المشروع)**
Organization's decision-makers, Sales Managers, Inventory Planners, and Marketing Teams.

**Stakeholders' Needs: (الاحتياجات المطلوب استيفائها في المشروع)**
They need to understand how different factors (like product categories and regional differences) impact overall sales performance. They also require reliable forecasts of future trends to optimize inventory levels and prevent stockouts.

**Functions & Features:** (المتطلبات التقنية)

- Documented analytical notebooks with clear SQL/Python code.
- Visualization plots detailing future sales forecasts.
- A comprehensive, interactive visualization dashboard combining all historical analysis and future predictions.

**Non-Functional Requirements: (متطلبات الجدوة والاداء)**

- **Usability:** The final Tableau dashboard must be intuitive, highly interactive, and easy to interpret for non-technical executives. سهولة الاستخدام
- **Performance:** The SQL queries and Pandas scripts must be optimized to efficiently process 10 years' worth of dataset records without significant performance bottlenecks. الاداء والكفاءة
- **Security & Tracking:** Ensure proper version control (e.g., using GitHub) to track project progress, manage code changes efficiently, and maintain the integrity of the analysis across different phases. امان المشروع